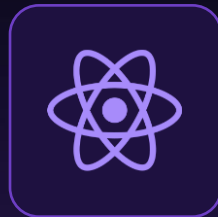


BTECH · MERN STACK · LECTURE 1

THE MERN STACK

One language. Front to back. MongoDB · Express · React · Node



Why JavaScript runs the whole show

Why is JavaScript the backbone?



FRONTEND — in the Browser

JavaScript is the **ONLY** language every browser natively understands — it makes web pages interactive: clicks, animations, live updates (React runs here).



Runs on: Chrome's V8 engine, Firefox, Safari...



BACKEND — on the Server

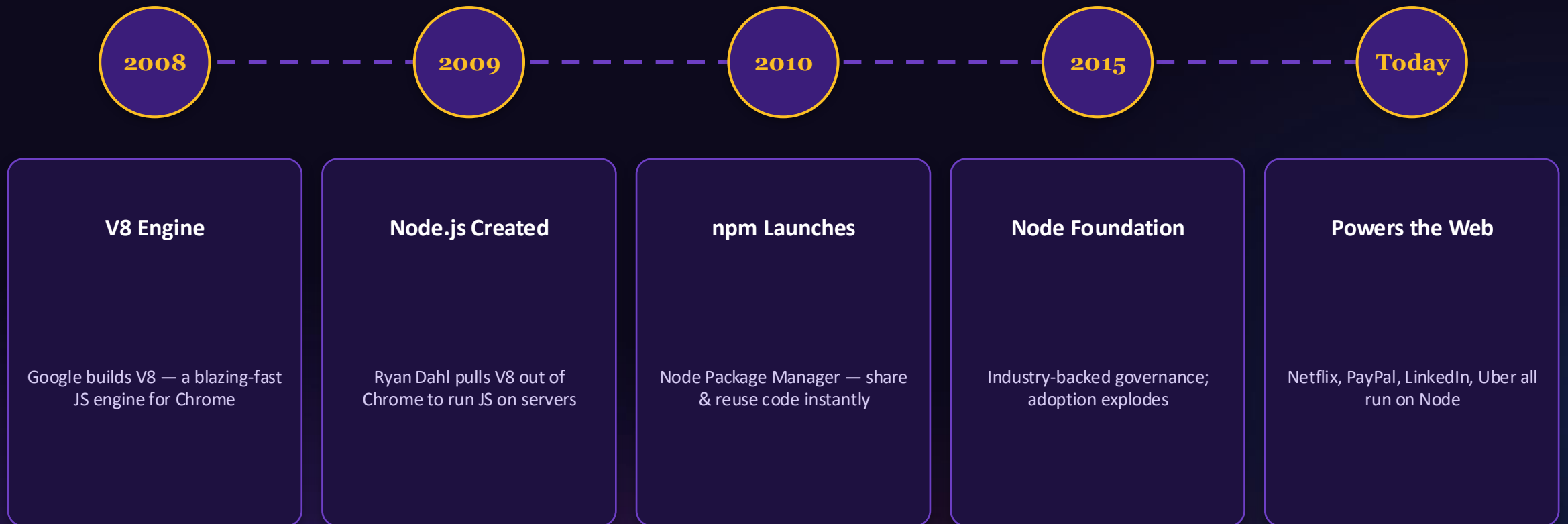
Node.js took Chrome's V8 engine **OUT** of the browser — so the same JavaScript can now read files, talk to databases, and run servers (Node + Express run here).



Runs on: Node.js runtime (server, your laptop)

*One language, **zero context-switching.***

How Node.js was born



From one engineer's idea to the backbone of the modern internet.

Node isn't the only runtime in town



Deno

By Node's original creator — secure by default, built-in TypeScript



Bun

Extremely fast all-in-one runtime, bundler & package manager



Python (Django)

Popular backend alternative — different language, same server role



PHP (Laravel)

Powers a huge share of the web, incl. WordPress



Ruby on Rails

Convention-over-configuration backend framework



Go (Golang)

Built for speed & concurrency at massive scale

We choose Node because it keeps you in **one language: JavaScript.**

Node gives power. Express gives structure.



RAW NODE.JS

- ✗ You manually parse every URL
- ✗ You write routing logic from scratch
- ✗ No built-in middleware system
- ✗ Verbose code for simple tasks

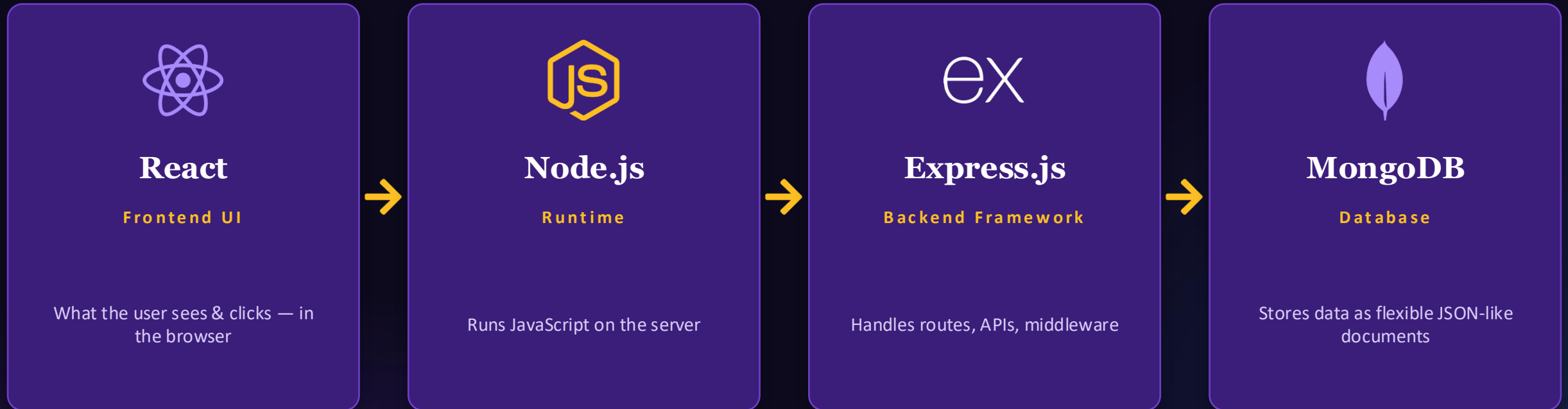


EXPRESS.JS FRAMEWORK

- ✓ Clean routing: `app.get()`, `app.post()`
- ✓ Middleware pipeline for auth, logging, JSON
- ✓ Easy REST API creation
- ✓ Huge plugin ecosystem via npm

Express turns Node from a raw engine into a ready-to-drive car — that's what makes it a framework, not just a library.

The full picture — how MERN fits together



React talks to Express, Express talks to MongoDB, Node runs it all — one language end to end.

This is the stack you will build real projects on, this semester.

Your official references — bookmark these

developer.mozilla.org



MDN Web Docs

The gold standard for JavaScript & web fundamentals

nodejs.org



nodejs.org

Official Node.js docs, downloads & API reference

expressjs.com



expressjs.com

Official Express guide, routing & middleware docs

react.dev



react.dev

The new official React docs — learn hooks & components

mongodb.com/docs



mongodb.com

Official MongoDB manual & free MongoDB University

npmjs.com



npmjs.com

Search & explore every package you will ever install

WHAT'S NEXT

Your MERN Journey Starts Now



JS Foundations



Node.js Core



Express APIs



MongoDB



React Frontend

One language. Four technologies. One connected skill for your entire career.

See you in Lecture 2 — JavaScript Foundations Refresher